

# Sun, Heat & the Strand

*UV damage, thermal styling, and the single habit that protects more than any product you will ever sell*

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## Learning Objectives

- Explain how UV exposure degrades hair, and why colour-treated hair is especially vulnerable
- Explain what physically happens to a strand above its critical heat threshold
- Recommend a safe styling temperature range by hair type
- Explain how a heat protectant works, and why it is never optional before thermal styling

Two very different kinds of damage, one careless afternoon

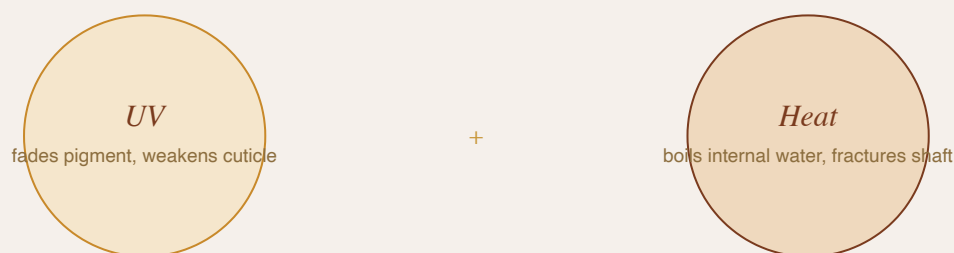
A client who spends a day at the beach and then blow-dries and flat-irons her hair that evening has, in a matter of hours, delivered her strands two entirely different assaults — one from the sun, one from her own styling tools. Both matter. Both are almost entirely preventable. This module treats them together because, in practice, they compound each other: sun-weakened hair is far more vulnerable to heat damage than hair that has been protected.

What UV actually does to a strand

Unlike skin, hair has no living cells to repair themselves — once a strand leaves the follicle, any damage to it is permanent until it is cut away or grown out. UV radiation breaks down the melanin in the cortex through a process called photo-oxidation, which is why hair visibly lightens and can take on a brassy or reddish cast after prolonged sun exposure — bleaching from the inside out, with no SPF bottle able to stop it entirely. UV also degrades the proteins in the cuticle, weakening the same disulfide bonds you studied in Module One and leaving the

surface rougher, duller, and more porous. Colour-treated hair is especially vulnerable, because the cuticle has already been chemically opened once and has less structural reserve to withstand a second assault.

◆ Figure 7.1 · Two Kinds of Damage



Sun degrades the strand from within over time; heat can fracture it in a single pass. Sun-weakened hair tolerates far less heat.

The heat threshold, and what "bubble hair" means

Hair holds a small amount of water within the cortex even when it looks and feels dry. Applied heat above roughly 450°F (232°C) can bring that internal moisture to a boil, forming tiny steam bubbles inside the shaft that fracture the strand from within — visible under magnification as what practitioners call bubble hair, and invisible to the naked eye until the strand simply snaps. This is why "the highest setting gets the job done faster" is one of the more expensive assumptions a stylist can make.

Safe styling temperatures by type

- **Fine hair:** 250–300°F (120–150°C) — fine strands reach their damage threshold at lower temperatures and rarely need more.
- **Medium hair:** 300–350°F (150–175°C) — sufficient for most styling needs without excess risk.
- **Coarse or tightly coiled hair:** 350–400°F (175–205°C) at most, and only with a heat protectant — coarser textures can tolerate slightly more heat, but the ceiling still exists.

Regardless of type, the working principle is the same: use the lowest temperature that achieves the result, and never repeat a pass over the same section more than once or twice.

The single habit that matters more than any product

Just as Course I taught that daily SPF protects more than any serum on a client's shelf, this course teaches its equivalent for hair: **never apply direct heat to hair that has not been protected first**. A heat protectant works by coating the strand with a thin, heat-resistant film — often silicone- or polymer-based — that raises the temperature at which real damage begins and helps distribute heat more evenly across the strand, reducing hot spots. It does not make hair heatproof. It buys margin. Teach every client this one non-negotiable: protectant first, tool second, always, no exceptions for "just a quick touch-up."

## Key Terms

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[object Object]

1

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2

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## Practical Activity · The Protectant Habit

For one week, apply a heat protectant before every single instance of heat styling on your own hair — no exceptions, including "quick" touch-ups. Note in a short journal whether this changed your styling habits, and what, if anything, you noticed about your hair by the end of the week.